

M5 Sales Data Processing Report

Preparation of a clean, model-ready dataset for 28-day retail sales forecasting

Item	Value
Source dataset	M5 Forecasting — Accuracy (Walmart)
Raw dataset location	raw_dataset/ (preserved, unmodified)
Processed dataset location	processed_dataset/sales_long_full.parquet
Processed rows	59,181,090
Processed columns	22
Storage format	Apache Parquet (snappy compression)
Storage size	287.09 MB
Forecasting features created	None — deferred to next stage

1. Purpose

This stage exists to turn the raw, competition-formatted M5 files into a single, clean, analysis-ready table before the team begins exploratory analysis and model development. The M5 sales data ships in a wide, one-column-per-day layout that is convenient for storage but awkward for time-series modeling libraries, and it is split across three separate files (sales, calendar, prices) that must be joined correctly before any forecasting work can begin.

The output of this stage — a single long-format table with sales, calendar context, and price already joined — will be used later as the starting point for exploratory data analysis (EDA), feature engineering (e.g. lag features, rolling statistics), and training a 28-day-ahead demand forecasting model. No modeling or feature-engineering decisions were made in this stage; the goal here is strictly a trustworthy, well-documented dataset.

2. Raw Dataset

The original M5 files were located in **DATASET_m5-forecasting-accuracy/** and were renamed (contents untouched) to **raw_dataset/** so the processed output could live alongside it without any risk of the source files being overwritten.

File	Rows	Columns	Size
calendar.csv	1,969	14	101 KB
sales_train_evaluation.csv	30,490	1,947	116 MB
sales_train_validation.csv	30,490	1,919	114 MB
sell_prices.csv	6,841,121	4	194 MB
sample_submission.csv	60,980	29	5 MB

Key characteristics: 3,049 unique products across 10 stores in 3 US states (CA, TX, WI), spanning 1,969 calendar days (2011-01-29 to 2016-06-19). The evaluation file contains the full observed sales history, `d_1` through `d_1941` (1,941 days), and was confirmed identical to the validation file on every day they share.

Preservation confirmation

MD5 checksums of all 5 raw files were captured before renaming the folder and re-verified after all processing completed. Every file matched exactly — the raw dataset was never modified.

File	Checksum verified unchanged
calendar.csv	Yes
sales_train_evaluation.csv	Yes
sales_train_validation.csv	Yes
sample_submission.csv	Yes
sell_prices.csv	Yes

3. Data Quality Audit

Every raw file was audited directly (not assumed from documentation) for missing values, duplicate records, invalid or negative values, date consistency, ID consistency, price consistency, and the relationships between tables (sales day identifiers vs. calendar, store/item IDs vs. price records, week identifiers vs. price coverage).

Check	Finding	Action taken
Missing sales values	0	None needed
Negative sales values	0	None needed
Duplicate sales IDs	0	None needed
Missing / duplicate calendar dates	0 / 0	None needed
Duplicate calendar 'd' identifiers	0	None needed
Event name/type mismatches	0	None needed
Missing prices in sell_prices.csv	0	None needed
Negative / zero prices	0 / 0	None needed
Duplicate store-item-week price rows	0	None needed
Sales 'd' columns missing from calendar	0	None needed
Store/Item ID mismatch (sales vs. prices)	0 / 0	None needed
Rows with no matching price after join	12,299,413 (20.78%)	Left as missing (see Section 8)

Conclusion: the raw dataset contains no genuine defects. No values required fixing. This matches the earlier exploratory analysis already stored in **analysis_output/**.

4. Data Transformation

The processed table was built through the following pipeline:



Before — raw wide format (excerpt)

id	item_id	store_id	d_1	d_2	d_3
HOBBIES_1_001_CA_1_evaluation	HOBBIES_1_001	CA_1	0	0	0

After — long format, joined (excerpt)

id	item_id	store_id	d	date	sales
HOBBIES_1_001_CA_1_evaluation	HOBBIES_1_001	CA_1	d_1	2011-01-29	0
HOBBIES_1_001_CA_1_evaluation	HOBBIES_1_001	CA_1	d_2	2011-01-30	0
HOBBIES_1_001_CA_1_evaluation	HOBBIES_1_001	CA_1	d_3	2011-01-31	0

30,490 series × 1,941 days = **59,181,090 rows**. The row count stayed identical before and after each join, confirming no accidental duplication occurred.

5. Processed Dataset Structure

File: **processed_dataset/sales_long_full.parquet** — 22 columns, described below.

Column	Type	Meaning / modeling value
id	category	Original M5 series identifier (item+store)
item_id / dept_id / cat_id	category	Product hierarchy — enables grouped/hierarchical modeling
store_id / state_id	category	Location hierarchy — enables store- or region-level modeling

Column	Type	Meaning / modeling value
d	string	Original day identifier, traceable back to raw files
date	datetime64	Real calendar date — needed for any time-aware model
wm_yr_wk	int32	Walmart internal year-week key — the price join key
weekday / wday / month / year	category/int	Calendar context for seasonality
event_name_1/2, event_type_1/2	category	Holidays/events — future feature-engineering input
snap_CA / snap_TX / snap_WI	int8	SNAP benefit day flags — known demand driver in M5
sell_price	float32	Weekly selling price (nullable — see Section 8)
sales	uint16	Observed units sold — the forecasting target, unaltered from raw

6. Zero Sales

Zero-sales observations make up **68.0%** of all rows (40,241,819 of 59,181,090). Every one of these was preserved exactly.

"Zero-sales observations were preserved because they are valid observations and their underlying cause cannot be determined directly from the available dataset."

A zero can mean no customer wanted the product that day, that the item was temporarily out of stock, or something else entirely — the raw files give no direct signal to distinguish these cases. Guessing would inject an assumption the team has not agreed on, so zeros were left as recorded rather than removed, replaced, smoothed, or interpolated.

7. Sales Spikes

The highest single-day sales value observed is **763 units**. This and every other high value were kept exactly as recorded.

"Sales spikes were preserved because they may represent genuine demand behavior."

The dataset carries no explicit promotion labels, so there is no reliable way to tell a genuine demand surge apart from a promotional one. Automatically capping or removing large values would risk deleting real signal the forecasting model needs to learn from.

8. Price Data

Prices were joined onto the sales table using the composite key **store_id + item_id + wm_yr_wk**, matching each week's sales to that week's selling price at that store.

Metric	Value
Rows before/after join (should match)	59,181,090 / 59,181,090
Rows with a matched price	46,881,677
Rows with no matched price	12,299,413 (20.78%)
Duplicate store-item-week rows in sell_prices.csv	0

Missing prices were **not** filled in. Products enter a store's assortment at different points in time, so weeks before an item was stocked simply have no price record — inventing a value there would misrepresent history that never happened. The 20.78% missing rate was left as null (NaN) and documented so the modeling team can decide deliberately how to treat it later.

9. Calendar Data

Each row's **d** identifier was matched to **calendar.csv** on the **d** column, adding the real date plus weekday, month, year, week identifier, primary/secondary event name and type, and the three state-level SNAP flags.

This join was verified two ways: the row count stayed unchanged after joining (proving the calendar's **d** values are unique, so no rows were duplicated), and five sample days spanning the full date range — **d_1**, **d_500**, **d_1000**, **d_1500**, and **d_1941** — were checked individually against **calendar.csv** and all matched exactly.

This information matters for later modeling because retail demand is strongly seasonal and event-driven: holidays, SNAP benefit-distribution days, and day-of-week patterns are all known drivers of grocery and household sales in the M5 dataset.

10. What We Have Not Done Yet

This stage is deliberately limited to cleaning and structural preparation. It does **not** include:

- Forecasting model training
- Feature engineering (lag_1, lag_7, lag_28, rolling means/medians/std, days-since-last-sale, etc.)
- Smoothing, outlier removal, or imputing sales values
- Stockout or promotion labeling
- Model selection or hyperparameter tuning
- The final 28-day prediction
- Any decision about the project's novelty or modeling strategy

These are all future stages the team will design and decide on together.

11. Validation

After processing, an independent validation script re-read the raw CSVs from scratch (rather than trusting the build script's own numbers) and cross-checked the output.

Check	Result
Row count = 30,490 x 1,941	59,181,090 (match)
item_id / store_id / id sets match raw	Yes / Yes / Yes
Sum of all sales matches raw	66,927,173 = 66,927,173
Zero-sales count matches raw	Match
Min / max sales match raw	0 / 763 (match)
5 random (id, day) sales values spot-checked	5 / 5 matched
5 calendar dates spot-checked (d_1...d_1941)	5 / 5 matched
5 random prices spot-checked	5 / 5 matched
Duplicate (id, d) pairs in processed table	0
Latest day present is d_1941 (no future days)	Confirmed
Raw file checksums unchanged (5 files)	5 / 5 identical

12. Final Dataset

Property	Value
Location	processed_dataset/sales_long_full.parquet
Format	Apache Parquet (pyarrow engine, snappy compression)
Rows	59,181,090
Columns	22
Size on disk	287.09 MB

Property	Value
Main purpose	Clean, joined, long-format base table for future EDA and feature engineering

13. Next Step

"Next stage: Exploratory analysis and feature-engineering design, followed by baseline forecasting and evaluation. The forecasting strategy and project novelty will be selected collaboratively by the team."